complete java class code example of each combination of intermediate and terminal operations

filter() boolean checks + findfirst() conditional check before first match

filter() null checks + findfirst() conditional check before first match

filter() comparisons + findfirst() conditional check before first match

filter() method calls returning boolean + findfirst() conditional check before first match

distinct() rely on equals() / hashcode() + findfirst() conditional check before first match

distinct() immutable objects + findfirst() conditional check before first match

map() transform objects + findfirst() conditional check before first match

map() create new objects + findfirst() conditional check before first match

map() call pure functions + findfirst() conditional check before first match

flatmap() return stream + findfirst() conditional check before first match

flatmap() split data + findfirst() conditional check before first match

flatmap() expand collections + findfirst() conditional check before first match

sorted() natural ordering + findfirst() conditional check before first match

sorted() custom comparator + findfirst() conditional check before first match

sorted() multi-field compare + findfirst() conditional check before first match

peek() logging + findfirst() conditional check before first match

peek() debugging + findfirst() conditional check before first match

peek() metrics + findfirst() conditional check before first match

Below are **all 18 programs converted to Java 1.5 compatible code**.

Since Java 1.5:

- ❌ No Stream API
- ❌ No Optional
- ❌ No lambda expressions
- ❌ No method references
- ❌ No List.of(), Map.of()
- ❌ No Objects.hash()
- ❌ No Comparator.comparing()

We replace everything with:

- ✔ for-each loops
- ✔ Iterator
- ✔ Collections / Arrays
- ✔ HashSet for distinct
- ✔ Manual sorting with Comparator
- ✔ Null checks
- ✔ Primitive counters instead of AtomicInteger

Below are **complete, standalone Java class examples** for **each requested Stream combination**, where:

- **Terminal operation = findFirst()**
- Purpose = **conditional check before the first matching element**
- Demonstrates **short-circuiting** (stream stops after first match)
- Java **8+ compatible**
- No collectors, no reduce, no extra logic

 findFirst() returns Optional<T> and respects **encounter order**.

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1 filter() boolean checks + findFirst()

```
import java.util.*;

public class FilterBooleanFindFirst {
   public static void main(String[] args) {

      List<Integer> list = Arrays.asList(11, 14, 18);
      Integer firstEven = null;

      for (Integer n : list) {
         if (n % 2 == 0) {
            firstEven = n;
            break;
         }
      }

      System.out.println(firstEven != null ? firstEven : -1);
   }
}
```

---

2 filter() null checks + findFirst()

```
import java.util.*;

public class FilterNullFindFirst {
   public static void main(String[] args) {

      List<String> list = Arrays.asList(null, "Anurag", "Amit");
      String firstNonNull = null;

      for (String s : list) {
         if (s != null) {
            firstNonNull = s;
            break;
         }
      }

      System.out.println(firstNonNull != null ? firstNonNull : "NONE");
   }
}
```

---

3 filter() comparisons + findFirst()

```
import java.util.*;

public class FilterComparisonFindFirst {
   public static void main(String[] args) {

      List<Integer> list = Arrays.asList(20000, 45000, 60000);
      Integer firstHigh = null;

      for (Integer s : list) {
         if (s >= 40000) {
            firstHigh = s;
            break;
         }
      }

      System.out.println(firstHigh != null ? firstHigh : 0);
   }
}
```

---

4 filter() method call returning boolean + findFirst()

```
import java.util.*;

public class FilterMethodFindFirst {

   static class User {
      String name;
      boolean active;

      User(String name, boolean active) {
         this.name = name;
         this.active = active;
      }

      boolean isActive() {
         return active;
      }
   }

   public static void main(String[] args) {

      List<User> list = Arrays.asList(
            new User("A", false),
            new User("B", true),
            new User("C", true)
      );

      User firstActive = null;

      for (User u : list) {
         if (u.isActive()) {
            firstActive = u;
            break;
         }
      }

      System.out.println(firstActive != null ? firstActive.name : "NONE");
   }
}
```

---

5 distinct() rely on equals() / hashCode() + findFirst()

```
import java.util.*;

public class DistinctEqualsFindFirst {

   static class Person {
      String name;

      Person(String name) {
         this.name = name;
      }

      public boolean equals(Object o) {
         if (o instanceof Person) {
            Person p = (Person) o;
            return name.equals(p.name);
         }
         return false;
      }

      public int hashCode() {
         return name.hashCode();
      }
   }

   public static void main(String[] args) {

      List<Person> list = Arrays.asList(
            new Person("Anurag"),
            new Person("Anurag"),
            new Person("Ravi")
      );

      Set<Person> set = new LinkedHashSet<Person>(list);

      Person first = null;
      for (Person p : set) {
         first = p;
         break;
      }

      System.out.println(first != null ? first.name : "NONE");
   }
}
```

6 distinct() immutable objects + findFirst()

```
import java.util.*;

public class DistinctImmutableFindFirst {
   public static void main(String[] args) {

      List<String> list = Arrays.asList("Java", "Java", "Go");
      Set<String> set = new LinkedHashSet<String>(list);

      String first = null;
      for (String s : set) {
         first = s;
         break;
      }

      System.out.println(first != null ? first : "NONE");
   }
}
```

---

7 map() transform objects + findFirst()

```
import java.util.*;

public class MapTransformFindFirst {
   public static void main(String[] args) {

      List<String> list = Arrays.asList("anurag", "amit");
      String firstUpper = null;

      for (String s : list) {
         firstUpper = s.toUpperCase();
         break;
      }

      System.out.println(firstUpper != null ? firstUpper : "NONE");
   }
}
```

---

8 map() create new objects + findFirst()

```
import java.util.*;

public class MapCreateObjectFindFirst {

   static class UserDTO {
      String name;

      UserDTO(String name) {
         this.name = name;
      }
   }

   public static void main(String[] args) {

      List<String> list = Arrays.asList("Anurag", "Amit");
      UserDTO first = null;

      for (String s : list) {
         first = new UserDTO(s);
         break;
      }

      System.out.println(first != null ? first.name : "NONE");
   }
}
```

9 map() call pure functions + findFirst()

```
import java.util.*;

public class MapPureFunctionFindFirst {

   static int square(int n) {
      return n * n;
   }

   public static void main(String[] args) {

      List<Integer> list = Arrays.asList(2, 3, 4);
      Integer first = null;

      for (Integer n : list) {
         first = square(n);
         break;
      }

      System.out.println(first != null ? first : 0);
   }
}
```

---

10 flatMap() return Stream + findFirst()

```
import java.util.*;

public class FlatMapStreamFindFirst {
   public static void main(String[] args) {

      List<List<Integer>> list = new ArrayList<List<Integer>>();
      list.add(Arrays.asList(10, 20));
      list.add(Arrays.asList(30));

      Integer first = null;

      for (List<Integer> inner : list) {
         for (Integer n : inner) {
            first = n;
            break;
         }
         if (first != null) break;
      }

      System.out.println(first != null ? first : 0);
```

```
    }
}
```

---

11 flatMap() split data + findFirst()

```
import java.util.*;

public class FlatMapSplitFindFirst {
    public static void main(String[] args) {

        List<String> list = Arrays.asList("Java Stream", "Flat Map");
        String firstWord = null;

        for (String s : list) {
            String[] words = s.split(" ");
            for (int i = 0; i < words.length; i++) {
                firstWord = words[i];
                break;
            }
            if (firstWord != null) break;
        }

        System.out.println(firstWord != null ? firstWord : "NONE");
    }
}
```

---

12 flatMap() expand collections + findFirst()

```
import java.util.*;

public class FlatMapExpandFindFirst {
   public static void main(String[] args) {

      Map<String, List<String>> map = new HashMap<String, List<String>>();
      map.put("A", Arrays.asList("1", "2"));
      map.put("B", Arrays.asList("3"));

      String first = null;

      for (List<String> values : map.values()) {
         for (String v : values) {
            first = v;
            break;
         }
         if (first != null) break;
      }

      System.out.println(first != null ? first : "NONE");
   }
}
```

---

13 sorted() natural ordering + findFirst()

```
import java.util.*;

public class SortedNaturalFindFirst {
   public static void main(String[] args) {

      List<Integer> list = new ArrayList<Integer>(Arrays.asList(5, 1, 3));
      Collections.sort(list);

      Integer first = list.size() > 0 ? list.get(0) : null;

      System.out.println(first != null ? first : 0);
   }
}
```

---

14 sorted() custom comparator + findFirst()

```
import java.util.*;

public class SortedCustomFindFirst {
   public static void main(String[] args) {

      List<String> list = new ArrayList<String>(Arrays.asList("Anurag", "Ravi", "Amit"));

      Collections.sort(list, new Comparator<String>() {
         public int compare(String a, String b) {
            return b.compareTo(a);
         }
      });

      String first = list.size() > 0 ? list.get(0) : null;

      System.out.println(first != null ? first : "NONE");
   }
}
```

---

15 sorted() multi-field compare + findFirst()

```
import java.util.*;

public class SortedMultiFieldFindFirst {

   static class Employee {
      String name;
      int age;

      Employee(String name, int age) {
         this.name = name;
         this.age = age;
      }
   }

   public static void main(String[] args) {

      List<Employee> list = new ArrayList<Employee>();
      list.add(new Employee("Anurag", 30));
      list.add(new Employee("Anurag", 25));
      list.add(new Employee("Ravi", 35));

      Collections.sort(list, new Comparator<Employee>() {
         public int compare(Employee e1, Employee e2) {
            int nameCompare = e1.name.compareTo(e2.name);
            if (nameCompare != 0) return nameCompare;
            return e1.age - e2.age;
         }
      });

      Employee first = list.size() > 0 ? list.get(0) : null;
      System.out.println(first != null ? first.name + " " + first.age : "NONE");
   }
}
```

---

16 peek() logging + findFirst()

```
import java.util.*;

public class PeekLoggingFindFirst {
   public static void main(String[] args) {

      List<Integer> list = Arrays.asList(1, 2, 3);
      Integer first = null;

      for (Integer n : list) {
         System.out.println("Peek: " + n);
         first = n;
         break;
      }

      System.out.println(first != null ? first : 0);
   }
}
```

---

17 peek() debugging + findFirst()

```
import java.util.*;

public class PeekDebugFindFirst {
   public static void main(String[] args) {

      List<Integer> list = Arrays.asList(10, 20, 30);
      Integer first = null;

      for (Integer n : list) {
         if (n == 20) {
            System.out.println("Debug hit");
         }
         first = n;
         break;
      }

      System.out.println(first != null ? first : 0);
   }
}
```

---

18 peek() metrics + findFirst()

```
import java.util.*;

public class PeekMetricsFindFirst {
   public static void main(String[] args) {

      List<Integer> list = Arrays.asList(5, 10, 15);
      int counter = 0;
      Integer first = null;

      for (Integer n : list) {
         counter++;
         first = n;
         break;
      }

      System.out.println("First: " + (first != null ? first : 0));
      System.out.println("Processed elements: " + counter);
   }
}
```